

Class 3:

Evaluate the following integrals by changing to polar coordinates:

1. $\int_0^a \int_y^a \frac{xdxdy}{x^2 + y^2}$

ans: $\frac{\pi a}{4}$

2. $\int_0^a \int_y^a \frac{x^2}{\sqrt{x^2 + y^2}} dxdy$

ans: $\frac{a^3}{3} \log(\sqrt{2} + 1)$
